

1. You have developed a tool (questionnaire) to assess health behaviors among adolescents of high school age. Responses to items in the questionnaire are used to calculate a risk index for each child: a number from 0 to 40 in which 0=very low risk and 40=extremely high risk. You want to assess health behaviors of adolescents in Indiana, and you therefore select 12 high schools where students will be administered your questionnaire. You believe that high schools will differ overall in the health behavior of their students. Use the dataset “school” to answer the questions below.

- a) You want to test your hypothesis that high schools will differ in the health behavior of their students. In your analysis, you will treat high school as a random effect. Why is this a better choice than as a fixed effect? (1 point)
- b) What is one consideration as you choose your 12 high schools for data collection? (1 point)
- c) You aren't able to collect data from all kids in each selected school, so you select 40 kids/school. What is one consideration as you choose these 40 kids in each school? (1 point)
- d) Suppose τ_i is the effect of the i th school on an adolescent's risk index (compared with an overall mean, μ). Write out your statistical model, explaining all notation and assumptions. (3 points)
- e) What are your null and alternative hypotheses as you test for differences among high schools? (2 points)
- f) What is your conclusion from this statistical test? How do you know? Include your SAS code here. (4 points)
- g) What are your estimates for σ^2 and σ_τ^2 ? (2 points)

1. You have developed a tool (questionnaire) to assess health behaviors among adolescents of high school age. Responses to items in the questionnaire are used to calculate a risk index for each child: a number from 0 to 40 in which 0=very low risk and 40=extremely high risk. You want to assess health behaviors of adolescents in Indiana, and you therefore select 12 high schools where students will be administered your questionnaire. You believe that high schools will differ overall in the health behavior of their students. Use the dataset “school” to answer the questions below.

- a) You want to test your hypothesis that high schools will differ in the health behavior of their students. In your analysis, you will treat high school as a random effect. Why is this a better choice than as a fixed effect? (1 point)

We are not interested in these 12 specific schools. Instead, they should be randomly chosen from all high schools in Indiana, so they can represent all high schools in the state.

- b) What is one consideration as you choose your 12 high schools for data collection? (1 point)

You want to randomly select your schools from a sampling frame (a list) of all high schools in the state. Where do you get such a list? In that list, do you include public schools only, or others (private, religious, charter, etc. ... what about kids who are homeschooled, have dropped out of school, or are adolescents but are not yet in high school?) – all these choices will determine the population to which you can draw inference.

- c) You aren’t able to collect data from all kids in each selected school, so you select 40 kids/school. What is one consideration as you choose these 40 kids in each school? (1 point)

Again, you want to randomly select kids. You first need a list of all kids who attend that school. Where can you get an up-to-date list? What privacy concerns or permissions do you need to address first? Do you want to take a simple random sample from the list, or first randomly sample classrooms, then randomly sample kids? How will you handle kids with low attendance or kids who may not be able to complete a written questionnaire in English (kids learning English as a second language, kids with disabilities, etc.)? Are there adolescents who are not in high school? The answers to these questions will affect your analysis methods and the scope of your inference.

- d) Suppose τ_i is the effect of the i th school on an adolescent’s risk index (compared with an overall mean, μ). Write out your statistical model, explaining all notation and assumptions. (3 points)

$$Y_{ij} = \mu + \tau_i + \varepsilon_{ij}$$

Y_{ij} is the risk index for the j th child in the i th school

μ is average risk index of the population

τ_i is the effect of the i th school on a child’s risk index

ε_{ij} is the random error term for the j th child in the i th school

$i = 1, 2, \dots, 12$

$j = 1, 2, \dots, 40$

$$\varepsilon_{ij} \sim \text{iid } N(0, \sigma^2)$$

$$\tau_i \sim \text{iid } N(0, \sigma_\tau^2)$$

The ε_{ij} and τ_i are independent

- e) What are your null and alternative hypotheses as you test for differences among high schools? (2 points)

$$H_0 : \sigma_\tau^2 = 0$$

$$H_1 : \sigma_\tau^2 > 0$$

- f) What is your conclusion from this statistical test? How do you know? Include your SAS code here. (4 points)

We reject the null hypothesis: Indiana adolescents' risk index does vary among high schools in the state (p=0.0139).

```
proc mixed data=school cl covtest;
  class school;
  model behavior = ;
  random intercept / subject=school;
run;
```

- g) What are your estimates for σ^2 and σ_τ^2 ? (2 points)

$$\sigma^2: \quad 8.6200$$

$$\sigma_\tau^2: \quad 3.2551$$